NexAfrica Data Analysis Internship — TASK 1

Kindly Read the Instructions in this guide carefully

Retail Sales Analytics Dashboard

Project Level: Beginner → Intermediate
Project Type: Business Intelligence / Data Analytics
Duration: 5 weeks
Primary Tools: Excel OR Power BI OR SQL OR PYTHON
Optional: Tableau

NEXAFRICA

1. Project Overview

Project Title

Retail Sales Analytics Dashboard

Business Scenario

You are working as a Junior Data Analyst for a retail company. Management wants to understand how the business is performing across products, customers, regions, and time.

Your job is to analyze the Superstore Sales dataset and build an interactive dashboard that answers:

- How much revenue is the business generating?
- Which products and categories perform best?
- Which regions generate the most sales?
- Which customers contribute the most revenue?
- How are sales changing over time?
- Which products or areas need attention?
- What recommendations should management consider?

2. Dataset

Dataset

Superstore Sales Dataset —Download here (Kaggle)

The dataset should contain fields similar to:

Field	Description
Order ID	Unique order identifier
Order Date	Date the order was placed
Ship Date	Date the order was shipped
Customer ID	Unique customer identifier
Customer Name	Customer name
Segment	Customer segment
Country	Customer country
City	Customer city
State	Customer state
Region	Sales region
Product ID	Product identifier
Category	Product category
Sub-Category	Product sub-category
Product Name	Product name
Sales	Revenue generated
Quantity	Number of units sold
Discount	Discount applied
Profit	Profit generated

Important: The exact columns may differ depending on the Superstore version you download. Interns should inspect the dataset before beginning the analysis.

NEXAFRICA

3. Project Objectives

By completing this project, interns should demonstrate their ability to:

1. Import and inspect a real-world dataset.
2. Clean and transform data.
3. Calculate business KPIs.
4. Analyze sales trends.
5. Analyze customer behavior.
6. Analyze product performance.
7. Perform customer segmentation.
8. Perform basic cohort analysis.
9. Build an interactive dashboard.
10. Extract business insights.
11. Make data-driven recommendations.

12. Present findings to a non-technical audience.

4. Tools — Choose any tool you are comfortable

Option A — Excel

Use:

- Excel Tables
- Power Query
- PivotTables
- PivotCharts
- Slicers
- Conditional Formatting
- Formulas

Useful functions:

- SUM
- SUMIFS
- COUNT
- COUNTIFS
- AVERAGE
- XLOOKUP
- IF
- YEAR
- MONTH
- TEXT
- UNIQUE
- FILTER

Option B — Power BI

Use:

- Power Query
- Data Model
- Relationships
- DAX
- Measures
- Cards
- Charts

NEXAFRICA

- Slicers
- Drill-down
- Interactive dashboard

Recommended DAX concepts:

- SUM
- COUNT
- DISTINCTCOUNT
- DIVIDE
- CALCULATE
- FILTER
- Time intelligence

Option C — SQL

Use SQL for:

- Data exploration
- Data cleaning
- KPI calculations
- Aggregation
- Customer analysis
- Product analysis
- Regional analysis
- Monthly trends

Recommended SQL concepts:

- SELECT
- WHERE
- GROUP BY
- ORDER BY
- HAVING
- CASE WHEN
- COUNT
- COUNT DISTINCT
- SUM
- AVG
- MIN
- MAX
- JOIN
- CTE

NEXAFRICA

Option D — Python for Data Analysis

Interns using Python should complete the analysis in Jupyter Notebook, Google Colab, or VS Code.

Recommended Libraries

- **Pandas** — data cleaning and manipulation
- **NumPy** — numerical operations
- **Matplotlib** — data visualization
- **Seaborn** — statistical visualization
- **Plotly** — interactive charts and dashboards

WEEK 1 — DATA PREPARATION & BUSINESS UNDERSTANDING

Objective

Understand the dataset, clean the data, and prepare it for analysis.

Step 1: Understand the Business Problem

Before touching the data, write a short explanation answering:

What is the purpose of this analysis?

Example:

Business Objective:
Analyze retail sales performance, customer purchasing behavior, and product performance to identify trends, opportunities, and areas requiring management attention.

Step 2: Inspect the Dataset

Determine:

- Number of rows
- Number of columns

NEXAFRICA

- Column names
- Data types
- Date range
- Number of customers
- Number of orders
- Number of products
- Number of categories
- Number of regions

Create a simple **Data Dictionary**.

Example:

Column	Data Type	Description
Order ID	Text	Unique order
Order Date	Date	Order date
Customer ID	Text	Unique customer
Sales	Decimal	Revenue
Quantity	Integer	Units sold
Profit	Decimal	Profit
Category	Text	Product category

Step 3: Data Cleaning

Check for:

Missing Values

Identify:

- Blank customer IDs
- Missing product names
- Missing sales
- Missing dates
- Missing profit

Duplicates

Check whether duplicate records exist.

NEXAFRICA

Incorrect Data Types

Make sure:

- Dates → Date
- Sales → Number
- Profit → Number
- Quantity → Integer
- Discount → Decimal

Data Consistency

Check:

- Category names
- Region names
- Product names
- Customer names

Step 4: Create Useful Calculated Fields

Depending on your tool, create fields such as:

Year

Extract the year from Order Date.

Month

Extract the month.

Year-Month

Create a field such as:

2025-Jan

This will help with monthly trend analysis.

Profit Margin

Conceptually:

Profit Margin = Profit ÷ Sales × 100

NEXAFRICA

Average Order Value

AOV = Total Sales ÷ Number of Orders

Week 1 Expected Deliverables

Submit:

1. Raw dataset
2. Cleaned dataset
3. Data dictionary
4. Data-cleaning documentation
5. Business problem statement
6. List of calculated fields

WEEK 2 — SALES & KPI ANALYSIS

Objective

Analyze overall business performance and identify sales trends.

Step 1: Calculate Core KPIs

Your dashboard should contain at least:

KPI 1 — Total Sales

Total revenue generated.

KPI 2 — Total Profit

Total profit generated.

KPI 3 — Total Orders

Number of unique orders.

KPI 4 — Total Customers

Number of unique customers.

NEXAFRICA

KPI 5 — Total Quantity Sold

Total units sold.

KPI 6 — Profit Margin

Percentage of sales retained as profit.

KPI 7 — Average Order Value

Average revenue per order.

Step 2: Analyze Sales Over Time

Analyze:

- Sales by year
- Sales by month
- Sales growth
- Profit by month
- Quantity by month

Create a **monthly sales trend chart**.

Questions:

1. Which month had the highest sales?
2. Which month had the lowest sales?
3. Are sales increasing or decreasing?
4. Are there seasonal patterns?

Step 3: Regional Analysis

Analyze:

- Sales by region
- Profit by region
- Orders by region
- Profit margin by region

Questions:

NEXAFRICA

Which region generates the most sales?

Which region generates the most profit?

Is the region with the highest sales also the most profitable?

Step 4: Category Analysis

Analyze:

- Sales by category
- Profit by category
- Quantity by category

Example:

Technology vs Furniture vs Office Supplies

Week 2 Expected Deliverables

Submit:

- KPI calculations
- Sales trend analysis
- Regional analysis
- Category analysis
- At least **6 visualizations**
- Minimum **5 business insights**

WEEK 3 — PRODUCT & CUSTOMER ANALYSIS

Objective

Understand which products and customers drive business performance.

PART A — PRODUCT ANALYSIS

Analyze:

Top 10 Products by Sales

NEXAFRICA

Identify the products generating the most revenue.

Top 10 Products by Profit

Identify the most profitable products.

Bottom 10 Products by Profit

Identify products generating low or negative profit.

Sub-Category Analysis

Analyze:

- Sales
- Profit
- Quantity

by sub-category.

Questions to Answer

1. Which product generates the most sales?
2. Which product generates the most profit?
3. Which products have negative profit?
4. Which category performs best?
5. Which sub-category needs attention?

PART B — CUSTOMER ANALYSIS

Calculate:

- Total customers
- Sales per customer
- Orders per customer
- Profit per customer
- Average customer order value

Customer Segmentation

NEXAFRICA

Segment customers based on purchasing behavior.

A simple segmentation approach:

High-Value Customers

Customers with high sales and profit contribution.

Medium-Value Customers

Customers with moderate purchasing activity.

Low-Value Customers

Customers with relatively low sales contribution.

You can create your own thresholds, but document the logic used.

Week 3 Expected Deliverables

Submit:

- Top 10 products analysis
- Bottom 10 products analysis
- Category/sub-category analysis
- Customer analysis
- Customer segmentation
- At least 5 additional insights

WEEK 4 — COHORT ANALYSIS & ADVANCED BUSINESS ANALYSIS

Objective

Analyze customer purchasing behavior over time and identify customer retention patterns.

Part A — Cohort Analysis

A cohort groups customers based on the period when they first purchased.

NEXAFRICA

For example:

Customer	First Purchase	Cohort
Customer A	Jan 2024	Jan 2024
Customer B	Feb 2024	Feb 2024
Customer C	Jan 2024	Jan 2024

Then analyze whether those customers returned in later months.

Questions

1. How many new customers joined each month?
2. Which cohort generated the most revenue?
3. Which cohorts have stronger repeat purchasing behavior?
4. How does customer activity change after the first purchase?

For beginners, a simple monthly cohort table is sufficient.

Part B — Discount Analysis

Analyze the relationship between:

Discount → Sales → Profit

Questions:

- Which products receive the highest discounts?
- Does higher discounting increase sales?
- Does higher discounting reduce profit?
- Which categories are most affected?

This is an important business-analysis component.

Part C — Profitability Analysis

Identify:

- High-sales/high-profit products
- High-sales/low-profit products

NEXAFRICA

- Low-sales/high-profit products
- Low-sales/low-profit products

This helps management distinguish revenue performance from actual profitability.

Week 4 Expected Deliverables

Submit:

- Cohort analysis
- Discount analysis
- Profitability analysis
- Customer retention insights
- At least 5 business findings
- Recommendations based on the findings

WEEK 5 — INTERACTIVE DASHBOARD DEVELOPMENT

Objective

Combine the analysis into a professional interactive dashboard.

Dashboard Page 1 — Executive Overview

Include:

KPI Cards

- Total Sales
- Total Profit
- Total Orders
- Total Customers
- Quantity Sold
- Profit Margin
- Average Order Value

Charts

1. Sales Trend

NEXAFRICA

Monthly sales over time.

2. Sales by Region

Compare regional performance.

3. Sales by Category

Compare major product categories.

4. Profit by Category

Identify profitable and less profitable categories.

Dashboard Page 2 — Product Performance

Include:

- Top 10 products
- Bottom 10 products
- Sales by category
- Profit by category
- Sales by sub-category
- Profit by sub-category

Add filters for:

- Category
- Sub-category
- Region
- Year

Dashboard Page 3 — Customer Analysis

Include:

- Total customers
- Sales per customer
- Top customers
- Customer segments
- Customer sales distribution
- Cohort analysis

NEXAFRICA

Filters:

- Year
- Region
- Segment
- Customer

Dashboard Page 4 — Business Insights

Create a management-focused page containing:

Key Findings

Example structure:

Finding 1:
Technology generated the highest sales.

Finding 2:
A particular region generated strong sales but comparatively weak profit.

Finding 3:
Certain products produced negative profit.

Recommendations

Example:

Recommendation 1:
Review pricing and discount strategies for products with consistently low profit margins.

Important: These are examples only. Interns must generate recommendations from their own analysis.

FINAL EXECUTIVE SUMMARY

Prepare a 1–2 page executive summary.

NEXAFRICA

It should contain:

1. Business Problem

What was the company trying to understand?

2. Dataset

What data was analyzed?

3. Methodology

Explain:

- Data cleaning
- Transformation
- KPI analysis
- Customer segmentation
- Cohort analysis
- Dashboard development

4. Key Findings

Provide 5–10 important findings.

5. Recommendations

Provide 5 actionable recommendations.

6. Conclusion

Summarize the overall business performance.

FINAL DASHBOARD REQUIREMENTS

The completed dashboard should be:

Interactive

Users should be able to filter the analysis.

Professional

Use:

NEXAFRICA

- Consistent fonts
- Clear titles
- Appropriate charts
- Proper spacing
- Consistent number formatting

Business-Focused

Avoid creating charts simply because they are available.

Every major visualization should answer a business question

NEXAFRICA

FINAL PROJECT SUBMISSION

Each intern should submit a folder structured approximately like this:

```
Customer_Churn_Project/
├── 01_Raw_Data/
│   └── superstore_raw.csv
├── 02_Cleaned_Data/
│   └── superstore_cleaned.xlsx
├── 03_Analysis/
│   └── analysis.xlsx / SQL_queries.sql
├── 04_Dashboard/
│   └── retail_sales_dashboard.pbix
├── 05_Report/
│   └── executive_summary.pdf
└── 06_Presentation/
```

└── final_presentation.pptx

For Excel-only submissions, replace the .pbix file with the final .xlsx dashboard.

FINAL ASSESSMENT RUBRIC

Area	**Weight**
Data Cleaning & Preparation	15%
KPI & Sales Analysis	15%
Customer Analysis & Segmentation	15%
Product Analysis	10%
Cohort & Advanced Analysis	10%
Dashboard Design & Interactivity	20%
Business Insights & Recommendations	10%
Documentation & Presentation	5%
Total	**100%**

Project Workflow

RAW DATA → CLEAN DATA → ANALYSIS → KPIs → CUSTOMER ANALYSIS → PRODUCT ANALYSIS → COHORT ANALYSIS → DASHBOARD → INSIGHTS → RECOMMENDATIONS

The final output should not simply be a collection of charts. It should function as a decision-support dashboard that allows a manager to understand sales performance, customer behavior, product performance, profitability, and areas requiring action.

NEXAFRICA